Topic: A=LU factorization

Question: Rewrite the matrix A in factored LU form.

$$A = \begin{bmatrix} 3 & 6 \\ -6 & 10 \end{bmatrix}$$

Answer choices:

A $\begin{bmatrix} 3 & 6 \\ -6 & 10 \end{bmatrix} = \begin{bmatrix} 3 & 6 \\ 0 & 10 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix}$

B $\begin{bmatrix} 3 & 6 \\ -6 & 10 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} 3 & 6 \\ 0 & 10 \end{bmatrix}$

C $\begin{bmatrix} 3 & 6 \\ -6 & 10 \end{bmatrix} = \begin{bmatrix} 3 & 6 \\ 0 & 22 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ -2 & 1 \end{bmatrix}$

D $\begin{bmatrix} 3 & 6 \\ -6 & 10 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ -2 & 1 \end{bmatrix} \begin{bmatrix} 3 & 6 \\ 0 & 22 \end{bmatrix}$

© Krista King Math

Solution: D

We'll apply an elimination matrix to A to zero-out the -6 in $A_{2,1}$.

$$E_{2,1}A = U$$

$$\begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} 3 & 6 \\ -6 & 10 \end{bmatrix} = \begin{bmatrix} 3 & 6 \\ 0 & 22 \end{bmatrix}$$

To solve this equation for A, we need to move the elimination matrix to the right side, inverting it.

$$\begin{bmatrix} 3 & 6 \\ -6 & 10 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix}^{-1} \begin{bmatrix} 3 & 6 \\ 0 & 22 \end{bmatrix}$$

To invert the elimination matrix, we'll change the sign on the non-zero entry below the main diagonal, and we'll get the $A = LU$ factorization.

$$\begin{bmatrix} 3 & 6 \\ -6 & 10 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ -2 & 1 \end{bmatrix} \begin{bmatrix} 3 & 6 \\ 0 & 22 \end{bmatrix}$$

© Krista King Math

Topic: A=LU factorization

Question: Find L if the matrix A is decomposed as LU.

$$A = \begin{bmatrix} 4 & 1 & -1 \\ 8 & 4 & -3 \\ 0 & -6 & 5 \end{bmatrix}$$

Answer choices:

A $L = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 2 & 1 \end{bmatrix}$

B $L = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 0 & -3 & 1 \end{bmatrix}$

C $L = \begin{bmatrix} 2 & 1 & -1 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{bmatrix}$

D $L = \begin{bmatrix} 4 & 1 & -1 \\ 0 & 2 & -1 \\ 0 & 0 & 2 \end{bmatrix}$

© Krista King Math

Solution: B

We'll apply an elimination matrix to A to zero-out the 8 in $A_{2,1}$.

$$E_{2,1}A = U$$

$$\begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 4 & 1 & -1 \\ 8 & 4 & -3 \\ 0 & -6 & 5 \end{bmatrix} = \begin{bmatrix} 4 & 1 & -1 \\ 0 & 2 & -1 \\ 0 & -6 & 5 \end{bmatrix}$$

Next we'll apply an elimination matrix to A to zero-out the -6 in $A_{3,2}$.

$$E_{3,2}E_{2,1}A = U$$

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 3 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 4 & 1 & -1 \\ 8 & 4 & -3 \\ 0 & -6 & 5 \end{bmatrix} = \begin{bmatrix} 4 & 1 & -1 \\ 0 & 2 & -1 \\ 0 & 0 & 2 \end{bmatrix}$$

To solve this equation for A, we need to move the elimination matrices to the right side, reversing their order and inverting each one.

$$\begin{bmatrix} 4 & 1 & -1 \\ 8 & 4 & -3 \\ 0 & -6 & 5 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}^{-1} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 3 & 1 \end{bmatrix}^{-1} \begin{bmatrix} 4 & 1 & -1 \\ 0 & 2 & -1 \\ 0 & 0 & 2 \end{bmatrix}$$

To invert the elimination matrices, we'll change the sign on the non-zero entry below the main diagonal, and we'll get the $A = LU$ factorization.

$$\begin{bmatrix} 4 & 1 & -1 \\ 8 & 4 & -3 \\ 0 & -6 & 5 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -3 & 1 \end{bmatrix} \begin{bmatrix} 4 & 1 & -1 \\ 0 & 2 & -1 \\ 0 & 0 & 2 \end{bmatrix}$$

Then we'll consolidate all the non-zero entries into one matrix, L.

© Krista King Math



$$\begin{bmatrix} 4 & 1 & -1 \\ 8 & 4 & -3 \\ 0 & -6 & 5 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 0 & -3 & 1 \end{bmatrix} \begin{bmatrix} 4 & 1 & -1 \\ 0 & 2 & -1 \\ 0 & 0 & 2 \end{bmatrix}$$

The question asked for the matrix L, which we found as

$$L = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 0 & -3 & 1 \end{bmatrix}$$

© Krista King Math

Topic: A=LU factorization

Question: Rewrite the matrix A in factored LDU form, where D is the diagonal matrix that factors the pivots out of U.

$$A = \begin{bmatrix} 3 & -3 & -3 & 6 \\ 12 & -10 & -12 & 26 \\ -6 & 8 & 8 & -6 \\ 6 & -6 & 4 & 38 \end{bmatrix}$$

Answer choices:

A $\begin{bmatrix} 1 & 0 & 0 & 0 \\ 4 & 1 & 0 & 0 \\ -2 & 1 & 1 & 0 \\ 2 & 0 & 5 & 1 \end{bmatrix} \begin{bmatrix} 3 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 6 \end{bmatrix} \begin{bmatrix} 1 & -1 & -1 & 2 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 1 \end{bmatrix}$

B $\begin{bmatrix} 1 & 0 & 0 & 0 \\ 4 & 1 & 0 & 0 \\ -2 & 1 & 1 & 0 \\ 2 & 0 & 5 & 1 \end{bmatrix} \begin{bmatrix} 3 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 6 \end{bmatrix} \begin{bmatrix} 3 & -3 & -3 & 6 \\ 0 & 2 & 0 & 2 \\ 0 & 0 & 2 & 4 \\ 0 & 0 & 0 & 6 \end{bmatrix}$

C $\begin{bmatrix} 1 & 0 & 0 & 0 \\ 4 & 1 & 0 & 0 \\ -2 & 1 & 1 & 0 \\ 2 & 0 & 5 & 1 \end{bmatrix} \begin{bmatrix} 1 & -1 & -1 & 2 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 1 \end{bmatrix}$

© Krista King Math

D $\begin{bmatrix} 1 & 0 & 0 & 0 \\ 4 & 1 & 0 & 0 \\ -2 & 1 & 1 & 0 \\ 2 & 0 & 5 & 1 \end{bmatrix} \begin{bmatrix} 3 & -3 & -3 & 6 \\ 0 & 2 & 0 & 2 \\ 0 & 0 & 2 & 4 \\ 0 & 0 & 0 & 6 \end{bmatrix}$

Solution: A

Let's set up $A = LU$.

$$\begin{bmatrix} 3 & -3 & -3 & 6 \\ 12 & -10 & -12 & 26 \\ -6 & 8 & 8 & -6 \\ 6 & -6 & 4 & 38 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ & 1 & 0 & 0 \\ & & 1 & 0 \\ & & & 1 \end{bmatrix} \begin{bmatrix} 3 & -3 & -3 & 6 \\ 12 & -10 & -12 & 26 \\ -6 & 8 & 8 & -6 \\ 6 & -6 & 4 & 38 \end{bmatrix}$$

We only need to zero-out the entries in U that are below the main diagonal. To zero-out the 12 in $U_{2,1}$, we need to subtract 4 of R_1 from R_2. Since that row operation is $R_2 - 4R_1$, we put 4 into $L_{2,1}$.

$$\begin{bmatrix} 3 & -3 & -3 & 6 \\ 12 & -10 & -12 & 26 \\ -6 & 8 & 8 & -6 \\ 6 & -6 & 4 & 38 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 4 & 1 & 0 & 0 \\ & & 1 & 0 \\ & & & 1 \end{bmatrix} \begin{bmatrix} 3 & -3 & -3 & 6 \\ 0 & 2 & 0 & 2 \\ -6 & 8 & 8 & -6 \\ 6 & -6 & 4 & 38 \end{bmatrix}$$

To zero-out the -6 in $U_{3,1}$, we need to add 2 of R_1 to R_3, which is the row operation $R_3 + 2R_1$. But we want this row operation written with subtraction, so we rewrite it as $R_3 - (-2)R_1$, and put -2 into $L_{3,1}$.

© Krista King Math

$$\begin{bmatrix} 3 & -3 & -3 & 6 \\ 12 & -10 & -12 & 26 \\ -6 & 8 & 8 & -6 \\ 6 & -6 & 4 & 38 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 4 & 1 & 0 & 0 \\ -2 & & 1 & 0 \\ & & & 1 \end{bmatrix} \begin{bmatrix} 3 & -3 & -3 & 6 \\ 0 & 2 & 0 & 2 \\ 0 & 2 & 2 & 6 \\ 6 & -6 & 4 & 38 \end{bmatrix}$$

To zero-out the 6 in $U_{4,1}$, we need to subtract 2 of R_1 from R_4. Since the row operation is $R_4 - 2R_1$, we put 2 into $L_{4,1}$.

$$\begin{bmatrix} 3 & -3 & -3 & 6 \\ 12 & -10 & -12 & 26 \\ -6 & 8 & 8 & -6 \\ 6 & -6 & 4 & 38 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 4 & 1 & 0 & 0 \\ -2 & & 1 & 0 \\ 2 & & & 1 \end{bmatrix} \begin{bmatrix} 3 & -3 & -3 & 6 \\ 0 & 2 & 0 & 2 \\ 0 & 2 & 2 & 6 \\ 0 & 0 & 10 & 26 \end{bmatrix}$$

To zero-out the 2 in $U_{3,2}$, we need to subtract 1 of R_2 from R_3. Since the row operation is $R_3 - R_2$, we put 1 into $L_{3,2}$.

$$\begin{bmatrix} 3 & -3 & -3 & 6 \\ 12 & -10 & -12 & 26 \\ -6 & 8 & 8 & -6 \\ 6 & -6 & 4 & 38 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 4 & 1 & 0 & 0 \\ -2 & 1 & 1 & 0 \\ 2 & & & 1 \end{bmatrix} \begin{bmatrix} 3 & -3 & -3 & 6 \\ 0 & 2 & 0 & 2 \\ 0 & 0 & 2 & 4 \\ 0 & 0 & 10 & 26 \end{bmatrix}$$

We don't need to perform any row operation to keep the 0 in $L_{4,2}$, so

$$\begin{bmatrix} 3 & -3 & -3 & 6 \\ 12 & -10 & -12 & 26 \\ -6 & 8 & 8 & -6 \\ 6 & -6 & 4 & 38 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 4 & 1 & 0 & 0 \\ -2 & 1 & 1 & 0 \\ 2 & 0 & & 1 \end{bmatrix} \begin{bmatrix} 3 & -3 & -3 & 6 \\ 0 & 2 & 0 & 2 \\ 0 & 0 & 2 & 4 \\ 0 & 0 & 10 & 26 \end{bmatrix}$$

To zero-out the 10 in $U_{4,3}$, we need to subtract 5 of R_3 from R_4. The row operation is $R_4 - 5R_3$, so we put 5 into $L_{4,3}$.

© Krista King Math

$$\begin{bmatrix} 3 & -3 & -3 & 6 \\ 12 & -10 & -12 & 26 \\ -6 & 8 & 8 & -6 \\ 6 & -6 & 4 & 38 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 4 & 1 & 0 & 0 \\ -2 & 1 & 1 & 0 \\ 2 & 0 & 5 & 1 \end{bmatrix} \begin{bmatrix} 3 & -3 & -3 & 6 \\ 0 & 2 & 0 & 2 \\ 0 & 0 & 2 & 4 \\ 0 & 0 & 0 & 6 \end{bmatrix}$$

To rewrite the $A = LU$ factorization so that U has only 1s along its main diagonal, insert an identity matrix to rewrite the equation as $A = LIU$.

$$\begin{bmatrix} 3 & -3 & -3 & 6 \\ 12 & -10 & -12 & 26 \\ -6 & 8 & 8 & -6 \\ 6 & -6 & 4 & 38 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 4 & 1 & 0 & 0 \\ -2 & 1 & 1 & 0 \\ 2 & 0 & 5 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 3 & -3 & -3 & 6 \\ 0 & 2 & 0 & 2 \\ 0 & 0 & 2 & 4 \\ 0 & 0 & 0 & 6 \end{bmatrix}$$

We'll first pull out the 3 from $U_{1,1}$, dividing the first row of U by 3.

$$\begin{bmatrix} 3 & -3 & -3 & 6 \\ 12 & -10 & -12 & 26 \\ -6 & 8 & 8 & -6 \\ 6 & -6 & 4 & 38 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 4 & 1 & 0 & 0 \\ -2 & 1 & 1 & 0 \\ 2 & 0 & 5 & 1 \end{bmatrix} \begin{bmatrix} 3 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & -1 & -1 & 2 \\ 0 & 2 & 0 & 2 \\ 0 & 0 & 2 & 4 \\ 0 & 0 & 0 & 6 \end{bmatrix}$$

Then we'll pull out the 2 from $U_{2,2}$, dividing the second row of U by 2.

$$\begin{bmatrix} 3 & -3 & -3 & 6 \\ 12 & -10 & -12 & 26 \\ -6 & 8 & 8 & -6 \\ 6 & -6 & 4 & 38 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 4 & 1 & 0 & 0 \\ -2 & 1 & 1 & 0 \\ 2 & 0 & 5 & 1 \end{bmatrix} \begin{bmatrix} 3 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & -1 & -1 & 2 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 2 & 4 \\ 0 & 0 & 0 & 6 \end{bmatrix}$$

We'll pull out the 2 from $U_{3,3}$, dividing the third row of U by 2.

$$\begin{bmatrix} 3 & -3 & -3 & 6 \\ 12 & -10 & -12 & 26 \\ -6 & 8 & 8 & -6 \\ 6 & -6 & 4 & 38 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 4 & 1 & 0 & 0 \\ -2 & 1 & 1 & 0 \\ 2 & 0 & 5 & 1 \end{bmatrix} \begin{bmatrix} 3 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & -1 & -1 & 2 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 6 \end{bmatrix}$$

© Krista King Math

Finally, we'll pull out the 6 from $U_{4,4}$, dividing the fourth row of U by 4.

$$\begin{bmatrix} 3 & -3 & -3 & 6 \\ 12 & -10 & -12 & 26 \\ -6 & 8 & 8 & -6 \\ 6 & -6 & 4 & 38 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 4 & 1 & 0 & 0 \\ -2 & 1 & 1 & 0 \\ 2 & 0 & 5 & 1 \end{bmatrix} \begin{bmatrix} 3 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 6 \end{bmatrix} \begin{bmatrix} 1 & -1 & -1 & 2 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Now the matrix A is in factored LDU form, where D is the diagonal matrix that factors the pivots out of U, where L is a lower triangular matrix, U is an upper triangular matrix, and where both L and U have only 1s along their main diagonals.

© Krista King Math